

FacilityOS — Executive Platform Overview

Enterprise Facility Operations, Multi-Tenant Service Architecture & SLA Engine

Platform Name: FacilityOS Pro	Version: 2.0 Production
Production Link: facilityos-brown.vercel.app	GitHub Repo: Invarsoft/facilityos-pro
Architecture: Multi-Tenant SaaS	Deployment: Vercel Edge / Docker

1. Executive Summary

FacilityOS is an enterprise-grade, multi-tenant facility operations platform built to digitize maintenance workflows across universities, hospitals, corporate office parks, residential apartments, and commercial malls. By uniting ticket creation, specialist technician matching, mobile PWA capabilities, and verified OTP resolution into a single system, FacilityOS eliminates SLA breaches and manual helpdesk overhead.

2. System Features & Module Architecture

Module / Feature	Functional Capability	Business Impact
Multi-Tenant Hierarchy	Logical data isolation for 21+ facilities with custom tenant codes and branding.	100% data privacy & multi-campus scalability.
14 Trade Service Matrix	Dynamic activation of Plumbing, Electrical, HVAC, IT, Elevators, Carpentry, Sanitation, etc.	Custom service tailoring per facility type.
Specialist Match Engine	Ranks technicians by trade discipline, active workload, star rating, and location.	Eliminates misassigned jobs & lowers SLA breaches.
OTP Verification Flow	4-Digit Security PIN requirement before work order mark-as-done.	Prevents premature or fake ticket closures.
Categorized PWA Mobile App	Responsive bottom bar navigation with segmented tab filtering for phones.	Instant mobile accessibility for ground staff.
5-Level SLA Engine	Real-time countdown timers for Emergency (2h), Critical (4h), High (12h), Normal (24h).	Auto-escalations & flashing breach alerts.

3. Technical Architecture & Deployment Stack

- **Frontend:** Next.js 15 (App Router), React 19, TypeScript, Tailwind CSS
- **State Management:** React Context API with Multi-Tenant Scoping
- **Mobile Native:** Progressive Web App (PWA) with CapacitorJS (.ipa / .apk) Support
- **Cloud & Hosting:** Vercel Production Infrastructure & Docker Containerization
- **Security:** Role-Based Access Control (RBAC) & Security Audit Logging